DUAL-USE TECHNOLOGY & IP ASSET PRESENTATION

TRUSTIA AI

Dual-Use Level-4 Autonomous Driving & Tactical UGV Software Core
Deterministic C++20 / ROS 2 Humble & Multi-Platform CAN-FD Hardware Gateway

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Asking Price: $145,000 USD (All-Cash IP Buyout) / $35K OEM License

Title & Cap Table: 100% Sole Founder Owned & Zero Debt

Founder & Architect: Murat Furkan Bayram

Accreditations: STM Candidate Supplier • ASELSAN Supplier • EU Horizon

01 / Strategic Acquisition & Licensing Opportunity

COST-TO-REPLICATE VS. DIRECT INTEGRATION

Save 18-24 Months & $345K+ in Direct R&D

Developing a deterministic, ISO 26262 ASIL-D compliant Level-4 autonomy stack from scratch demands **18 to 24 months** and over **$345,000 USD** in engineering payroll. Trustia AI provides an instant, production-verified alternative.

Pure Deterministic C++20 Engineering

Engineered with zero memory leaks, strict MISRA C++:2023 compliance, and deterministic execution loops, eliminating the random crashes and unbounded latencies common in academic ROS/Autoware setups.

Two Turnkey Transaction Pathways

Path A (Full Buyout): 100% assignment of all source code, algorithms, and IP via standard APA.
Path B (OEM Fleet License): Non-exclusive enterprise deployment for vehicle fleets ($35K + $1.5K/unit).

$145K
FULL BUYOUT ASKING PRICE

$345K+
MODELED REPLACEMENT COST

16,400+
DETERMINISTIC C++20 / PYTHON LOC

1,301
100% PASSED AUTOMATED TESTS

02 / Core Software & Algorithmic Architecture

DETERMINISTIC MODERN C++ (C++20) & ROS 2 HUMBLE

GNSS-Denied 3D NDT LiDAR SLAM

Centimeter-accuracy 3D Normal Distributions Transform (NDT) and Pose-Graph SLAM engine delivering robust localization in GPS-denied environments (forest canopies, tunnels, EW jamming zones, underground facilities).

Native CAN-FD Hardware Gateway

Direct drive-by-wire integration for electric trucks, passenger shuttles (Hyundai Ioniq 5 / Isuzu BIG.e / Novociti), and military UGV chassis. Injects steering (100Hz) and throttle/braking (50Hz) commands.

400Hz ESKF Multi-Sensor Fusion

Error-State Kalman Filter fusing 128-channel Ouster OS2-128 LiDAR, dual Livox Mid-360, 4D Radar, and wheel odometry at microsecond latency with zero sensor drift.

Kinematic Hybrid A* & MPC Path Control

Ackermann-constrained trajectory planning with sub-15ms cycle times, combined with Model Predictive Control (MPC) and Pure Pursuit for precision tracking.

03 / Dual-Use Defense & Tactical Capabilities

FROM CIVIL AUTONOMOUS TRANSPORT TO TACTICAL UNMANNED GROUND VEHICLES (UGV)

Tactical C2 Command Interface

Full desktop mission control console (tactical_gui.py) compliant with NATO STANAG 4586 and MIL-STD-2525 military standards with encrypted BVLOS teleoperation links.

Air-Ground Swarm Autonomy

Distributed coordination engine (air_ground_swarm.py) enabling autonomous task allocation, area search, and cooperative target tracking between UGVs and UAVs.

IED, Mine & CBRN Threat Detection

Multi-sensor fusion integrating GPR, magnetometer, and spectral sensors to detect and classify roadside IEDs, landmines, and hazardous threats in real time.

Built-in V2X Engine & Tactical Voice Command Integration

Includes production-ready v2x_engine.py for Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication, alongside voice_command.py for hands-free tactical operator tasking.

04 / Verification & Digital Twin Testing

1,301 AUTOMATED TESTS IN CARLA & GAZEBO DIGITAL TWIN

Subsystem / Test Suite	Test Count	Validation Engine	Status
Math Foundations & Coordinates	88 / 88	Deterministic CI/CD	100% PASS
3D NDT LiDAR SLAM & ESKF Fusion	73 / 73	Synthetic Point Clouds	100% PASS
Kinematic Hybrid A* & Ackermann	49 / 49	CARLA Urban Grid	100% PASS
CAN-FD Gateway & Actuator Control	33 / 33	Kvaser Virtual Bus	100% PASS
Perception, Obstacles & Classifiers	229 / 229	Gazebo & CARLA Sensors	100% PASS
Automotive Safety, Watchdog & E-Stop	28 / 28	ASIL-D Fault Injection	100% PASS
Stress Benchmarks & Scenario Matrix	734 / 734	End-to-End Simulation	100% PASS
Swarm, Threat & CBRN Modules	67 / 67	Tactical Loop Harness	100% PASS

Transparent Verification: Zero Critical Bugs

All 1,301 unit and integration tests have been successfully verified in high-fidelity CARLA and Gazebo digital twin simulation pipelines. Ready for track integration with hardware BOM specs.

ISO 26262 ASIL-D Functional Safety

- **5ms Human Override:** Instant torque-sensor preemption returning vehicle control to human driver.
- **200ms Independent Watchdog:** Dual-channel relay triggering hardware E-Stop upon computer fault.
- **Minimal Risk Maneuver (MRM):** Safe controlled stop in lane upon critical sensor loss.

05 / Institutional Accreditations & Credentials

DEFENSE, GOVERNMENT & EUROPEAN INNOVATION ECOSYSTEM

STM Defense Technologies Candidate Supplier

Officially recorded candidate supplier for STM Defense Technologies Engineering, specializing in unmanned tactical ground systems.

NATO NCAGE Military Supplier Code

Registered and approved military supplier code by Ministry of National Defense (NCAGE: TR26258467723).

ASELSAN Defense Approved Supplier

Officially registered and approved on ASELSAN Supplier Portal (Tracking No: 0050569) and deep-tech candidate.

TÜBİTAK ARBİS & KOSGEB Accreditations

National Researcher Registry (ARBİS: TBTK-0229-6571) and KOSGEB Advanced Entrepreneur Certificate (KSB01UGE0115153370).

European Commission Horizon Europe

Official Participant Identification Code (PIC: 861711529) and approved EIT Urban Mobility Partnership (ID: CUS15554).

ITO BTM Deep-Tech Venture Residency

Resident deep-tech venture at Istanbul Chamber of Commerce BTM Campus; member of Open Invention Network (OIN) patent pact.

06 / Transferred Asset Inventory (Turnkey IP)

COMPLETE DELIVERABLES TRANSFERRED TO BUYER / LICENSEE

1. Complete Proprietary Source Code

16,400+ lines of C++20 and Python codebase, ROS 2 Humble nodes, CAN-FD drivers, V2X engine, and full clean Git repository.

2. 100% Intellectual Property (IP) Transfer

Unconditional assignment of all SLAM algorithms, ESKF filters, kinematic planners, and system designs.

3. CARLA Digital Twin & 1,301 Test Suites

Automated CI/CD test harnesses, digital twin urban scenarios, and test reports proving 100% verification.

4. Hardware BOM Specifications & Wiring

Turnkey BOM ($32,800 USD kit) for Hyundai Ioniq 5, Isuzu BIG.e, and UGV platforms, sensor mounting CADs, and schematics.

5. Web Platform & Tactical C2 Console

Next.js 16 corporate platform, 3D teleoperation web client, STANAG 4586 desktop C2 interface, and primary domain.

6. 30 Days of Direct Founder Engineering Support

Hands-on architectural handover, code orientation, and integration guidance directly from founder Murat Furkan Bayram.

07 / Commercial Pathways & Direct Executive Contact

FLEXIBLE, LEGALLY SECURE ASSET BUYOUT OR OEM FLEET LICENSING

Transaction Options

Option A (Full Buyout): $145,000 USD (Turnkey 100% Asset Acquisition via Escrow)

Option B (OEM License): $35,000 USD Deployment + $1,500/vehicle Royalty

Legal Health: 100% Clean Title, Zero Debt, Zero Third-Party Claims

Closing Protocol: Mutual NDA → APA Execution → Escrow Deposit → 72h Code Handover

Integration Agility

The software stack is designed with modular Hardware Abstraction Layers (HAL), enabling rapid porting to commercial electric trucks, autonomous shuttles, or defense UGV platforms within 72 hours.

Direct Executive Contact

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DISCUSSIONS CONDUCTED UNDER MUTUAL NDA

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